

2-norm of a matrix

Asked 7 years, 10 months ago Modified 7 years, 10 months ago Viewed 6k times

Given that λ is an eigenvalue of $A^T A$ with eigenvector x I need to show

(1) $\|Ax\|_2^2 = \lambda \|x\|_2^2$

(2) $\|A\|_2 \leq \|A^T A\|_2^{\frac{1}{2}}$

I knew $\|Ax\|_2 = \sum_i \sum_j a_{ij} x_i$

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edited Feb 28, 2017 at 12:22

asked Feb 28, 2017 at 12:19

 Myshkin
36.6k 28 166 339

And what have you tried? Have you written out the explicit meaning of both sides of inequality 1, for example? Have you applied the fact that λ is an eigenvalue to try to compare these? – John Hughes Feb 28, 2017 at 12:22

What you say that "know" is wrong, as the example $A = [1]$, $x = [-1]$ shows. – John Hughes Feb 28, 2017 at 12:23

I see, then I must admit that I dont know the definition of this 2-norm of matrix – Myshkin Feb 28, 2017 at 12:24

please explain if you know .. – Myshkin Feb 28, 2017 at 12:24

$\|Ax\|$ is the 2-norm of a vector. – John Hughes Feb 28, 2017 at 13:01

1 Answer

Sorted by: Highest score (default)

The first point is proven as follows.

From the SVD of $A = UDV^T$ we can see that eigenvalues of $A^T A = VD^2V^T$ are just squared ones from A . At the same time the columns of V are the eigenvectors of $A^T A$. So, exploiting orthogonality of eigenvectors

$$\|Ax\|_2^2 = \|UDVx\|_2^2 = \|D(Vx)\|_2^2 = \|De_\lambda \|x\|_2\|_2^2 = \|\sqrt{\lambda} \|x\|_2\|_2^2 = \lambda \|x\|_2^2$$

The proof is based on the property of the second matrix norm

$$\|A\|_2 = \sqrt{\lambda_{\max}(A^T A)}$$

$$\|A^T A\|_2 = \sqrt{\lambda_{\max}(A^T A A^T A)}.$$

The same reasoning from the first point says that the eigenvalues of $A^T A A^T A$ are just the ones of $A^T A$ being squared. So

$$\begin{aligned} \|A\|_2 &= \sqrt{\lambda_{\max}(A^T A)} \\ \|A^T A\|_2 &= \sqrt{\lambda_{\max}(A^T A A^T A)} = \sqrt{\lambda_{\max}^2(A^T A)} = |\lambda_{\max}(A^T A)| = \lambda_{\max}(A^T A) = \\ &= \max(\lambda_{\min}^2(A), \lambda_{\max}^2(A)) \geq \|A\|_2^2 \end{aligned}$$

Edit

The inequality becomes after the notion that $\lambda_{\max}(A^T A) = \max(\lambda_{\min}^2(A), \lambda_{\max}^2(A))$, because eigenvalues of A could be negative.

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edited Feb 28, 2017 at 12:51

answered Feb 28, 2017 at 12:38

 Andrei Kulunchakov
2,462 12 20

Here's a simpler answer for the first one: $\|Ax\|_2^2 = \|\lambda x\|_2^2 = \lambda^2 \sum x_i^2 = \lambda^2 \|x\|_2^2$, so taking square roots, $\|Ax\|_2 = |\lambda| \|x\|_2$. (Note that the first claim is actually false, by the example I gave in my second comment on the questions, and that your proof implicitly assumes all evalues are nonnegative.) – John Hughes Feb 28, 2017 at 13:06